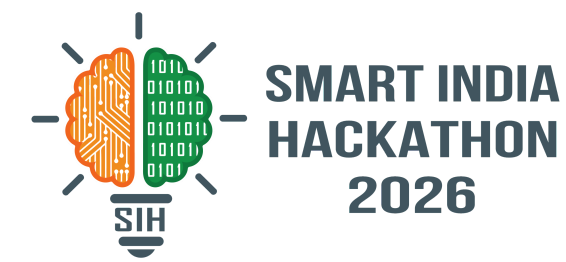


SMART INDIA HACKATHON 2026



- **Problem Statement ID –** SIH26106
- **Problem Statement Title-** AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**132753
- **Team Name -** Parallel Minds





Challenges v/s Solutions

Phishing & Impersonation Emails

Proposed Solution: AI/ML-based content analysis to detect phishing, BEC and impersonation attempts.

Hidden Technical Details in Emails

Proposed Solution: Automatic extraction and analysis of email headers (From, Reply-To, Received, etc.) with SPF, DKIM and DMARC checks.

Malicious URLs and Attachments

Proposed Solution: Automatic extraction and sandbox-based analysis of URLs and attachments using threat intelligence and reputation checks.

Unknown Origin of Emails

Proposed Solution: IP and domain analysis with geolocation mapping, ISP details and reputation check to trace email infrastructure.

Sender Identity Spoofing

Proposed Solution: Verify domain authenticity using SPF, DKIM, DMARC and detect sender/Reply-To mismatches.

Multiple Related Attacks (Campaigns)

Proposed Solution: Correlate indicators (IP, domain, URL, content) to detect similar emails and identify attack campaigns using graph-based analysis.

Lack of Actionable Information

Proposed Solution: Generate a clear risk score, explain findings and provide recommended actions for investigators.

Evidence Tampering & Lack of Trust

Proposed Solution: Blockchain-based immutable recording of email hashes, reports and investigator actions to ensure evidence integrity and audit trail.

Key Features

Email Upload & Analysis

Support .eml files with automatic parsing of content, headers, URLs, IPs and attachments.

AI Threat Detection

Detect phishing, BEC, spam and impersonation using AI/ML with explainable results.

Header & Domain Inspection

SPF, DKIM, DMARC verification and domain intelligence.

IP Geolocation & Threat Intel

Map source infrastructure and check against global threat intelligence sources.

URL & Attachment Analysis

Analyze URLs and attachments for malicious indicators using sandboxing and reputation data.

Attack Graph & Campaign Detection

Visualize relationships between emails, domains, IPs and URLs to detect attack campaigns.

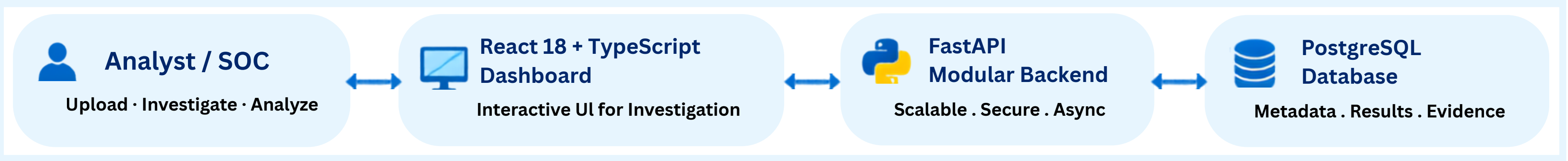
Forensic Reports & Case Management

Generate detailed reports, maintain case history, preserve evidence and manage investigation timeline.

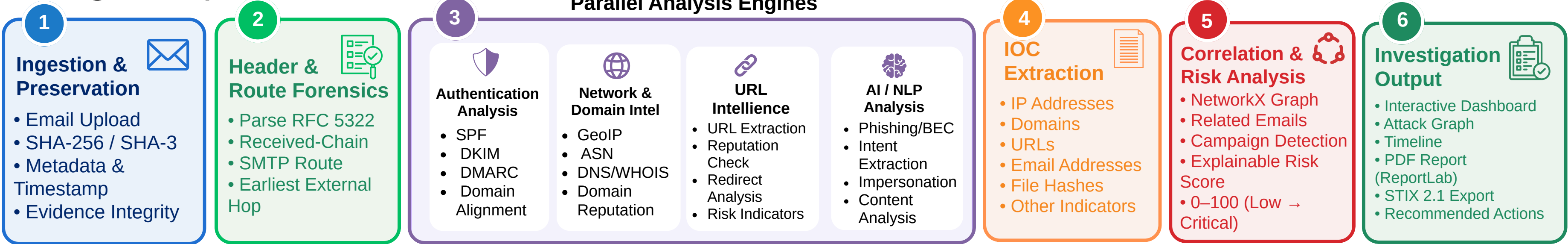
Blockchain-Based Evidence Integrity

Store hashes of emails, reports and key actions on a permissioned blockchain for a tamper-proof audit trail and chain of custody.

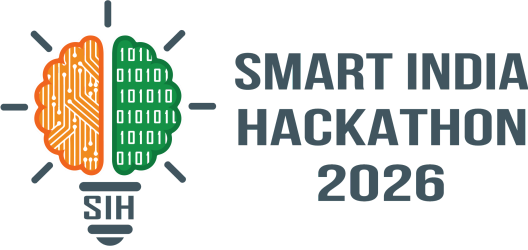
System Architecture



Investigation Pipeline



Parallel
mind



IMPACT AND BENEFITS

TARGET USERS



SOC / Security Analyst

Monitor, detect and respond faster



CERT / Incident Response Teams

Investigate and mitigate threats



Cybercrime Investigators

Monitor, detect and respond faster



Enterprise IT Security Teams

Monitor, detect and respond faster



Government & Educational Institutions

Secure communication in services



Digital Forensics Teams

Preserve and analyze evidence

KEY SECURITY IMPACTS



Faster Investigation

Reduce manual analysis of headers, domains, IPs and URLs.



Better Detection

Combine authentication, infrastructure, content and behavioral signals.



Campaign Visibility

Connect multiple suspicious emails through shared infrastructure and IOCs.



Evidence Preservation

Maintain hashes, timestamps and chain-of-custody records.

BENEFITS



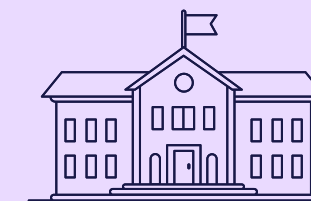
Social

Better protection against phishing and financial fraud.



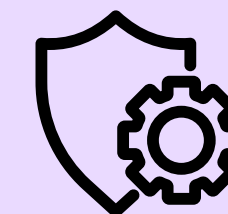
Economic

Reduce losses caused by BFC, credential theft and payment fraud.



Institutional

Improve incident investigation and reporting.



Cybersecurity

Convert isolated suspicious emails into campaign-level intelligence.

FEASIBILITY AND VIABILITY RESEARCH AND REFERENCES

FEASIBILITY

- Email parsing and security analysis can be implemented using established libraries and APIs.
- SPF/DKIM/DMARC, IP, DNS, and domain intelligence are technically achievable.
- AI classification and graph-based correlation can support automated threat detection.
- PostgreSQL and React provide a practical foundation for evidence storage and the SOC dashboard.

VIABILITY

- The MVP can work with public/synthetic datasets and modular threat-intelligence APIs.
- Hashing, RBAC, audit logs, and sandboxed analysis improve security and reliability.
- Explainable AI scoring with analyst review can manage false positives.
- The modular architecture allows future expansion into an enterprise SOC platform.

STANDARDS & TECHNICAL REFERENCES

-  **RFC 5322**
Internet Message Format
-  **RFC 7208 (SPF)**
RFC 6376 (DKIM)
RFC 7489 (DMARC)
-  **STIX 2.1**
Cyber Threat Intelligence Representation
-  **MITRE ATT&CK**
Threat behavior and adversary techniques

EXISTING SOLUTIONS STUDIED

-  **Microsoft Defender for Office 365**
Studied for enterprise phishing/BEC protection and email investigation capabilities.
-  **Google Messageheader**
Studied for SMTP header and routing analysis.
-  **VirusTotal**
Studied for URL, domain, file and IP reputation analysis.

KEY INSIGHTS & IDENTIFIED OPPORTUNITY

